Research Question

20 Years of Doppelter Pukelsheim in Switzerland: What Is the Mechanical Effect of the Transition from Hagenbach-Bischoff on the Seat Allocation of Smaller Parties and Party-System Fragmentation?

Hypotheses

H0₁: Across all analyzed elections, parties with less than 10% canton-wide vote share receive no more than 5% more seats under the double-proportional Pukelsheim system than they would have received under Hagenbach-Bischoff.

$$H_0: \frac{\sum Seats_{small,Pukelsheim}}{\sum Seats_{small,HB}} \leq 1.05$$

H1 (Main): Across all analyzed elections, parties with less than 10% canton-wide vote share receive more than 5% more seats under the double-proportional Pukelsheim system than they would have received under Hagenbach-Bischoff.

$$H_1: \frac{\sum Seats_{small,Pukelsheim}}{\sum Seats_{small,HB}} > 1.05$$

H0₂: Across the analyzed regions, party-system fragmentation did not increase between the earlier and later elections.

H2: Across the analyzed regions, party-system fragmentation increased between the earlier and later elections.

$$ENP = \frac{1}{\sum_{i=1}^{n} v_i^2}$$

v stands for vote shares of the parties

for control we also check the proportionality

$$ENP = \frac{1}{\sum_{i=1}^{n} s_i^2}$$

s stands for seat shares of the parties

Topic

Switzerland's cantonal electoral reforms of the last two decades have brought renewed attention to a central question of representative democracy: how parliamentary seats can be distributed in a way that reflects voter preferences as fairly as possible while still preserving territorial representation. In the Swiss case, this debate became especially prominent because traditional proportional systems in small constituencies often generated high implicit thresholds. These electoral systems therefore disadvantaged smaller parties which led to raising concerns about electoral equality (Bundeskanzlei, 2013, pp. 8–10). The double-proportional allocation method, commonly known as the double Pukelsheim, emerged in this context as a reform intended to reduce such distortions and to strengthen the equal weight of votes across the canton (Lüscher, 2023, p. 571).

At the core of this debate lies the contrast between two different logics of seat allocation. The traditional Hagenbach-Bischoff method belongs to the family of divisor methods with downward rounding and therefore tends to benefit larger parties, especially in smaller electoral districts. (Bundeskanzlei, 2013, pp. 6–7). By contrast, the double Pukelsheim combines a canton-wide upper allocation with a subsequent redistribution of seats to districts, thereby improving proportionality at the cantonal level and reducing the number of votes that remain without effective influence on seat allocation (Pukelsheim & Schuhmacher, 2011, pp. 1581–1582, 1591). In this sense, the reform is closely linked to the principle of Erfolgswertgleichheit, meaning that votes should contribute as equally as possible to the final electoral outcome (Bundeskanzlei, 2013, pp. 8–9).

This reform is especially relevant for the representation of smaller parties. Because the double Pukelsheim aggregates party strength across the entire canton before allocating seats to districts, votes that would previously have been ineffective in small constituencies can now contribute to a seat gain at the cantonal level (Lüscher, 2025, pp. 1–2). Sandro Lüscher illustrates this mechanical effect with the case of Schwyz in 2016: under the old Hagenbach-Bischoff system, the Green Liberal Party would not have won a single seat, whereas under the new method they obtained three seats (Lüscher, 2025, p. 2-3). More generally, recent work suggests that the shift away from Hagenbach-Bischoff reduces the structural advantage previously enjoyed by larger parties and improves the parliamentary chances of smaller competitors (Bundeskanzlei, 2025, pp. 22–25).
At the same time, the reform does not eliminate all tensions. While the double Pukelsheim improves party proportionality and vote equality, it may also produce new frictions at the district level. Scholars have pointed out that the system can weaken the direct correspondence between a district's seat share and its actual influence on the canton-wide outcome, thereby creating a tension between partisan proportionality and territorial representation (Lüscher, 2023, pp. 577–580).

Against this background, this thesis examines the mechanical effect of electoral reform on the parliamentary representation of smaller parties. More specifically, it asks and shows statistcally how many seats smaller parties gain through the transition from the Hagenbach-Bischoff system to the double Pukelsheim in Swiss cantonal parliamentary elections.

The central research gap of this thesis lies in the fact that the existing literature mainly discusses the Doppelproporz in legal, normative, or system-level terms: it emphasizes improved proportionality, reduced weightless votes, and better opportunities for smaller parties, while also pointing to trade-offs such as territorial distortions and counterintuitive seat allocations. However, there is still little systematic evidence on the mechanical seat effect of the reform itself: namely, how many seats smaller parties actually gained or would have lost when comparing the observed Pukelsheim allocation with a counterfactual Hagenbach-Bischoff allocation across elections and cantons. Recent work shows that smaller parties were generally better represented under the new system and that research has increasingly focused on broader institutional effects or territorial representation, but not on a standardized empirical measurement of seat gains for smaller parties across cases. The thesis fills this gap by translating the reform's abstract proportionality advantage into a concrete, comparable outcome measure: the net seat gain or loss of smaller parties under the new versus the old allocation rule.

Definition and Source of Data

Canton	Source	Data Type	Number of Periods | Start Year
Canton ZH	Archiv Wahlresultate Kantonsratswahlen	CSV	5 | 2007
Canton GR	Resultate Grosswahlen -Pressedaten	XLSX	1 |2022 (next 14.6.2026)
Canton VS	Archivierte -Abstimmungs und Wahlresultate Wahlresultate 2025	CSV/XLSX (divided in districts) 2025 only XLSX	3 | 2017
Canton SH	Abstimmungen und Wahlen - Resultate Results 2008,2012,2016	PDF (wrote email)	5 | 2008
Canton SZ	Kantonsratswahlen 2020,2024 Ergebnis 2016	CSV(2020) XLSX(2024)	3 | 2016

		PDF(2016) XLSX(2016) fragmented	
Canton AG	Statistisches Archiv	XLSX/CSV	5 | 2009
Canton NW	Landrat 2026/22 Result 2014 Result 2018	PDF(2022,18,14) WEB(2026) XLSX(14,18,22,26) through mail contact	4 | 2014
Canton ZG	Archiv Kantonsratswahlen	CSV (14,18,22) use Kandidierendenresultate (more comprehensive)	3 | 2014 (next 4.10.2026)
Canton UR	Mail	XLSX (20,24) über Mail	2 | 2020

Regarding the Results, which are only publicly available as PDFs: I contacted the affected cantons and asked for CSV/XLSX documents. Of whom Nidwalden and Schwyz already answered and sent me the data. If the canton of Schaffhausen can't provide me with data, then I'll ask the local newspapers ‚Schaffhauser Nachrichten and Schaffhauser AZ. Alternatively the SRG might have data. As a further source of Data also the Center for Democracy seems to have some sort of comprehensive but not public data. The federal statistical office BFS provides data on cantonal elections between 1919-2020 but misses results on district level. If none of these possibilities are working, I'd need to transform the data by myself into XLSX or CSV.

Units of Observations

Canton (Name)	Character
Election Year	Integer
Districts	Character
District_seats (seats available per district)	Integer
Party_votes_in district	Integer
Party_votes_cantonal_total	Integer
Seats_received_DP (Doppelter pukelsheim)	Integer
Simulated_seats_HB (Hagenbach Bischoff)	Integer
Share_of_votes (per party)	Double
Small_party	Boolean value
diff_hb_dp (difference in seats allocation | per party)	Integer

Method

The analysis takes the actual seat allocation under the Pukelsheimer Proporz as its starting point. Based on the available election data at the municipal or district level, it then reconstructs how many seats each party would have obtained under the former Hagenbach-Bischoff system. The comparison between the actual seat distribution under the new system and the hypothetical distribution under the old system makes it possible to identify how many seats each party gained or lost as a result of the reform. In this way, the study assesses the mechanical effect of the electoral-system change on party representation, with a particular focus on smaller parties.

For the purposes of this thesis, a smaller party is defined as a party that received at least 10% of the vote share in the relevant electoral unit. Youth lists and other special-interest lists, such as women's lists or pensioners' lists, are not treated as independent parties. Instead, their votes and seats are attributed to the respective mother party, which reflects common electoral practice in Switzerland.

The choice of 10% as the threshold is grounded in both legal and normative considerations. In its decision BGE 131 I 74, the Swiss Federal Supreme Court argued that direct and natural quorums, understood as barriers to representation within an electoral district, should generally not exceed 10%. The 2013 report of the Federal Chancellery refers to the same benchmark and further notes that the Venice Commission considers thresholds in the range of 10–15% to be the upper acceptable limit. Using vote share rather than absolute party size is particularly appropriate because a party may be electorally small in one region while being comparatively strong in another. Low thresholds are closely linked to the principle of *Erfolgswertgleichheit*, or equality of vote success value, which requires that votes should be translated into seats as equally and effectively as possible.

Investigating the threshold is especially important, since none of the reform cantons had a <10% threshold in all districts before the implantation of the new electoral reform (Lüscher,2024, p.115).

Presentation of Findings

To present the empirical findings, the thesis will use scatterplots at two levels of analysis. First, for each election, a scatterplot will show how many seats smaller parties actually received under the Pukelsheimer Proporz and how many seats they would have received under the Hagenbach-Bischoff system. Each smaller party will therefore be represented by two observations: one under the old system and one under the new system. The x-axis will display the number of seats, while the y-axis will indicate the party's vote share. This makes it possible to visualize whether parties with a given level of electoral support were systematically underrepresented or better represented depending on the electoral formula.

Second, these election-level results will be aggregated into an overall comparative scatterplot. In this summary figure, each election will be represented by two data points, one for the old system and one for the new system, combining the outcomes for all smaller parties within that election. This allows for an overall assessment of whether smaller parties benefited from the institutional reform across elections. In addition, the analysis may reveal regional differences, for example whether smaller parties in cantons such as Zurich benefited more strongly from the reform than smaller parties in cantons such as Graubünden. If the scope of the thesis permits, the analysis may also be extended to examine whether the effects vary systematically across specific parties rather than only across elections or cantons.

Literature Review

Bundeskanzlei,(2013) *: Proporzwahlsysteme im Vergleich; Bericht der Bundeskanzlei* i.C.w. BGE 131|74
This official federal document, is used to explain the methodology of the different election methods. Aswell as it's benefits and disadvantages.

Bundeskanzlei,(2025)*: Wahlsysteme und Sitzzuteilungsverfahren im Vergleich; Bericht der Bundeskanzlei.*
Shows the Historic Development of the electoral systems in Switzerland. Furthermore explains what the criterias are for electoral systems and shows the legal perspective. Contains numerical examples.

Fiva J., Olle F., (2014) *Mechanical and Psychological Effects of Electoral Reforms*
This paper studies the mechanical effects and can therefore give my further insight and inspiration into how to analyze these kind of mechanical effect and what to consider when doing so.

Lago I., I Coma F., (2026) *The 'odd–even effect': The link between the number of parties and district magnitude*
This paper also studies the mechanical effects and give me further insights.

Lüscher.S, (2023): *Parlamentswahlen nach dem doppelt-proportionalen Sitzzuteilungsverfahren im Lichte der territorialen Repräsentation. Eine empirische Annäherung an ein konzeptionelles Dilemma*
Focuses on "how the double-proportional seat allocation method affects constituency weighting and, consequently, the representation of the electorates and parties based in

those constituencies" Focuses on regions in which politcaly identity is strongly tied to district (e.g. Grisons).
This can be used for explaining of how the possible benefits of doppelter Pukelsheim, also have downsides. And to generally show the whole image of the discussion.

Lüscher S.,(2024) *Redeeming the Pledge of Electoral Equity? A Study of the Roads to and Effects of the Divisor-Based Biproportional Apportionment Method in Switzerland*
Dissertation at the UZH. The Dissertation is not publicly available, but I wrote an E-Mail to Mr. Lüscher. The Dissertation could possibly be suitable to compare with my calculations and findings.

Pukelsheim F., (2017): *Proportional Representation: Apportionment Methods and Their Applications*
This Book by the Mathemician behind the doppelter Pukelsheim Proporz, gives a good first source overview over the method and the reasoning behind it. It also shows the difficulties democracies face to take everything into account, when constructing an electoral system.

Pukelsheim F., Schuhmacer C., (2004) *Das neue Zürcher Zuteilungsverfahren für Parlamentswahlen*
This Paper can be used as a historical document since it explains the process of the inauguration of the new electoral system.

Pukelsheim F., Schuhmacher C., (2011) *Doppelproporz bei Parlamentswahlen – ein Aus- und Rückblick*
This paper analyses the Kantonratswahlen 2011 of the canton Zurich. It especially shows the difference in seat allocation between the 2 systems for this election in 2011. Therefore it comes quite close to what I want to do for all elections.